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# C++ Midway Project

## Description:

This program simulates a free-for-all multiplayer deathmatch. I originally designed it as a fun way to "draw names out of a hat," but eventually, this program will act as the framework for a simple role-playing game.

The program starts by prompting the user to input how many combatants are going to participate in the battle, then the user gives each combatant a name. Upon entering a name, the program assigns that combatant a randomly-generated number of hit points, a strength value, and a weapon.

Once all the names have been entered, the program generates a text file with a play-by-play description of the battle, consisting of funny verbs and body parts. Every turn, each combatant attacks every other combatant with a 50% chance to hit. If an attack hits, the damage done is determined by multiplying the combatant's strength value by their weapon damage. This continues until all but one of the combatants' hit points are less than or equal to zero.

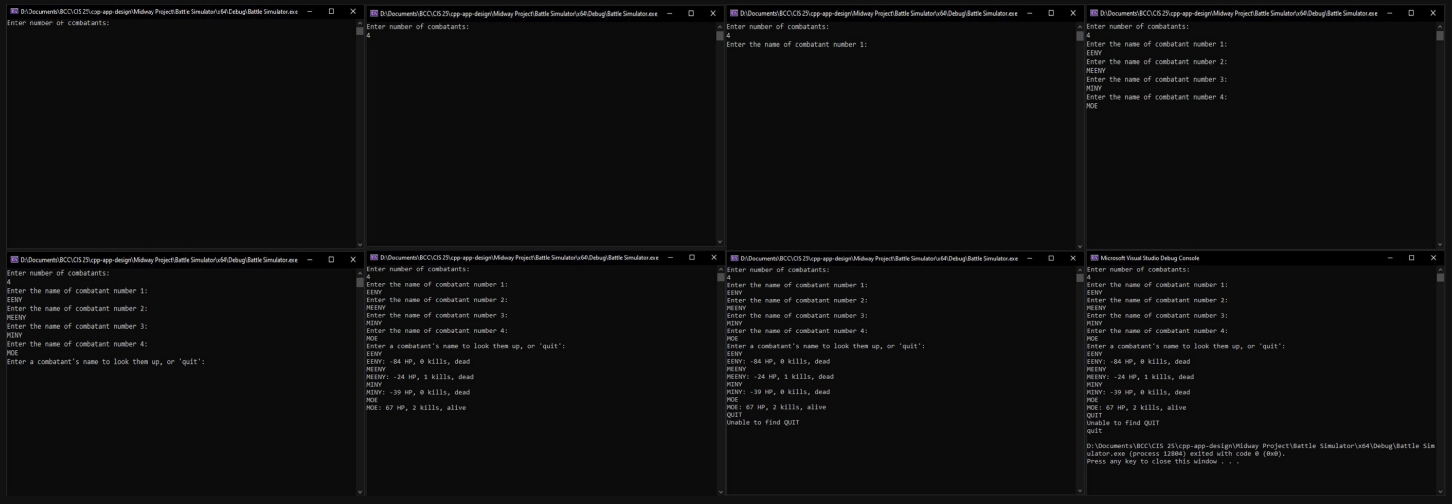
When the battle has concluded, the user is able to search for a name to see how they performed during the battle. This loop continues until the user enters 'quit.'

This program includes or demonstrates all of the requirements for the midway project:

- Data types and data sizes
- File types (e.g., .txt for data I/O)
- Use of **pointers**
- **arrays**
- A **binary search** algorithm (or any other search)
- Use of **strings**
- File I/O (read from or write to a file)
- At least **two classes**

However, even though this program includes multiple inherited classes, only the combatant class is currently functional. The functionality of the other classes will be implemented at a later date.

## Output Screenshots:



**Output Results:**

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1. EENY spawned with 180 HP and 20 strength.
2. MEENY spawned with 171 HP and 9 strength.
3. MINY spawned with 105 HP and 18 strength.
4. MOE spawned with 196 HP and 15 strength.

LET THE GAMES BEGIN!

1. EENY tried to attack MEENY, but they were distracted by a FLYING PIG!
2. EENY tried to attack MINY, but they were distracted by a FLYING PIG!
3. EENY tried to attack MOE, but they were distracted by a FLYING PIG!
4. MEENY cleaved EENY in the KIDNEY with a POINTY STICK
EENY took 9 points of damage!
EENY's remaining HP: 171
5. MEENY pelted MINY in the LEFT VENTRICLE with a piece of BOLOGNA
MINY took 9 points of damage!
MINY's remaining HP: 96
6. MEENY pelted MOE in the LUMBAR SPINE with a rubber DUCKY
MOE took 9 points of damage!
MOE's remaining HP: 187
7. MINY clobbered EENY in the RIGHT BUTTCHEEK with a rubber DUCKY
EENY took 18 points of damage!
EENY's remaining HP: 153
8. MINY cleaved MEENY in the MEDULLA OBLONGATA with a CHAINSAW
MEENY took 18 points of damage!
MEENY's remaining HP: 153
9. MINY tried to attack MOE, but they were distracted by a FLYING PIG!
10. MOE tried to attack EENY, but they were distracted by a FLYING PIG!
11. MOE tried to attack MEENY, but they were distracted by a FLYING PIG!
12. MOE clobbered MINY in the KIDNEY with a CHAINSAW
MINY took 165 points of damage!
MINY's remaining HP: -69
------< MOE killed MINY! >------
13. EENY tried to attack MEENY, but they were distracted by a FLYING PIG!
14. EENY pelted MOE in the STERNUM with a piece of BOLOGNA
MOE took 20 points of damage!
MOE's remaining HP: 167
15. MEENY cleaved EENY in the EYEBALL with a LIVE SQUID
EENY took 9 points of damage!
EENY's remaining HP: 144
16. MEENY stabbed MOE in the TRAPEZIUS with a TRASH CAN LID
MOE took 9 points of damage!
MOE's remaining HP: 158
17. MOE clobbered EENY in the CLAVICLE with a KATANA
EENY took 90 points of damage!
EENY's remaining HP: 54
18. MOE smacked MEENY in the LIVER with a CHAINSAW
MEENY took 120 points of damage!
MEENY's remaining HP: 33
19. EENY stabbed MEENY in the TRAPEZIUS with a WALRUS TOOTH
MEENY took 40 points of damage!
MEENY's remaining HP: -7
------< EENY killed MEENY! >------
20. EENY tried to attack MOE, but they were distracted by a FLYING PIG!
21. MOE impaled EENY in the MEDULLA OBLONGATA with a simple KITCHEN KNIFE
EENY took 30 points of damage!
EENY's remaining HP: 24
22. EENY whacked MOE in the LIVER with a ZWEIHANDER
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MOE took 60 points of damage!
MOE's remaining HP: 98
23. MOE clobbered EENY in the AORTA with a POINTY STICK
EENY took 45 points of damage!
EENY's remaining HP: -21
------< MOE killed EENY! >------
MOE is the last combatant standing!

   SCOREBOARD
1. MOE: 2 kills (SURVIVOR!)
2. EENY: 1 kills
3. MEENY: 0 kills
4. MINY: 0 kills